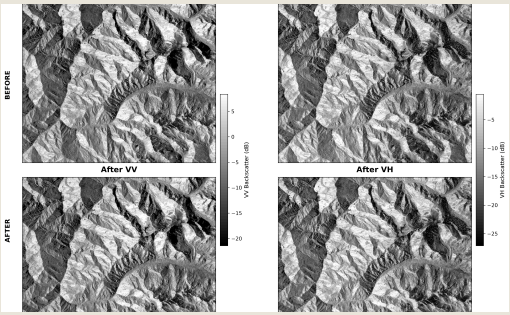
SATELLITE-BASED FLOOD IMPACT ASSESSMENT

Flood Intelligence: Satellite-Based Flood Impact Assessment

A rapid, evidence-led view of flood-related surface change, terrain pathways, infrastructure exposure, and network accessibility across northern Nepal.

EVENT · 26 AUGUST 2026 | **STUDY AREA** · RASUWA CORRIDOR, NORTHERN NEPAL

Decision lens: Read the results as a transparent screening and scenario assessment—not as a verified damage inventory or confirmed flood extent.



SENTINEL-1 BEFORE / AFTER EVIDENCE — 16 → 28 AUG 2026

VALIDATED SNAPSHOT · KEY FIGURES

529 BUILDINGS WITHIN HYDROLOGICAL CORRIDOR Corridor exposure records	**16** POTENTIALLY AFFECTED ROADS Mapped road records	**0** POTENTIALLY AFFECTED BRIDGES Current analysis criteria	**57** SETTLEMENTS ANALYSED Accessibility records
8 BASELINE-UNREACHABLE No pre-event route found	**12** POTENTIALLY CUT OFF Under simulated disruption	**18** LONGER ROUTES Under simulated disruption	**19** NO MEANINGFUL ROUTE CHANGE Under simulated disruption

ANALYTICAL WORKFLOW

01 / OBSERVE
Radar change evidence
Copernicus Sentinel-1 before/after surface-change signal.

02 / TRACE
Terrain & drainage
Copernicus DEM GLO-30 terrain structure and hydrological corridor context.

03 / OVERLAY
Infrastructure exposure
Pre-event OSM roads, bridges, buildings, settlements, and hospitals.

04 / ASSESS
Accessibility scenario
Road-network comparison after removing potentially affected segments.

INTERPRETATION GUARDRAILS

- Corridor exposure does not prove physical damage; potentially affected roads are not confirmed damaged roads.
- Cut-off and longer-route counts are simulated network-disruption scenarios, not confirmed real-world isolation.
- Zero potentially affected bridge records does not prove that no bridges were damaged.
- Sentinel-1 surface change is evidence, not a confirmed flood boundary.

DATA & ATTRIBUTION

- **Copernicus Sentinel-1** · radar evidence pair dated 16 and 28 August 2026.
- **Copernicus DEM GLO-30** · terrain and drainage context; source attribution retained for the Copernicus DEM product.
- **OpenStreetMap contributors** · pre-event infrastructure snapshot dated 27 July 2026.
- Snapshot: **rasuwa-0826-dashboard-v2** · 1.1 · generated 11 October 2026.

Contains modified Copernicus Sentinel data 2026. Copernicus DEM GLO-30 © DLR e.V. 2022. © OpenStreetMap contributors.